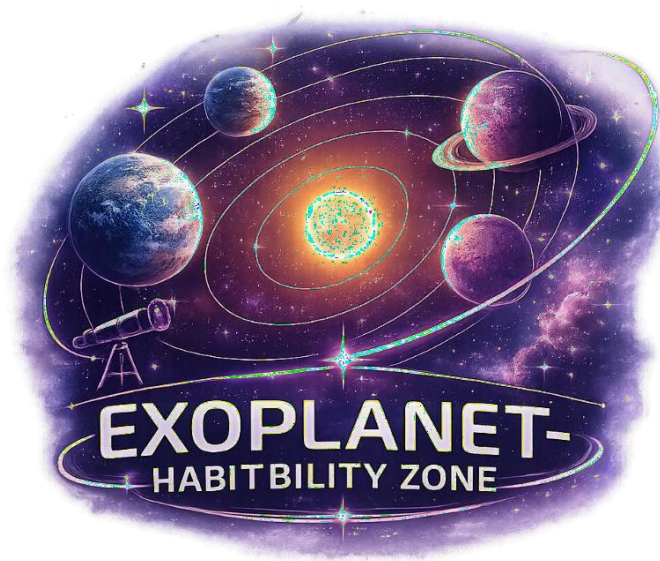


EXOPLANET HABITABILITY ZONE ANALYSIS

“Where Data Meets Discovery”



GRADUATION PROJECT DOCUMENTATION

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Chapter 1

Chapter 1 — Project Overview

1.1 Project Concept

Exoplanet Habitability Zone Analysis is a graduation project developed as part of the Digital Egypt Pioneers Initiative (DEPI), Data Analysis Track. The project uses real, continuously-updated observational data from NASA's Exoplanet Archive to identify which of the thousands of confirmed exoplanets fall inside their star's habitable zone (the "Goldilocks Zone") and to rank how physically similar each one is to Earth.

The analysis pipeline was built entirely with Python (pandas / NumPy) across a Medallion Architecture (Bronze → Silver → Gold), enriched with astrophysical calculations (stellar luminosity, received flux, equilibrium temperature, Earth Similarity Index), and delivered through an interactive Power BI dashboard plus a machine learning model that predicts habitability for a hypothetical planet.

The motivation behind this project is twofold:

- **Analytical:** apply the full data analysis lifecycle — ingestion, cleaning, scientific feature engineering, dimensional modeling, and interactive visualization — to a real, publicly maintained astrophysics dataset.
- **Educational:** translate raw astronomical measurements (flux, equilibrium temperature, orbital mechanics) into an accessible tool that lets a general audience explore which known exoplanets could plausibly support life.

Project objectives span five domains:

- **Data Engineering:** ingest, validate, and clean a real NASA exoplanet dataset (~6,100 planets, 19 raw columns) using Python across the Medallion Architecture layers.
- **Feature Engineering:** derive scientific properties that are not directly observed — stellar luminosity (Stefan–Boltzmann law), received flux (inverse-square law), planet equilibrium temperature, escape velocity, and the Earth Similarity Index (ESI).
- **Dimensional Modeling:** design and populate a star-schema data model in the Gold layer — one fact table and four dimension tables — optimized for Power BI.
- **Dashboard Development:** build a multi-page interactive Power BI dashboard covering discovery history, physical/orbital characteristics, stellar properties, and habitability ranking.
- **Predictive Modeling:** train a habitability classifier and an Earth Similarity regressor, and expose them through an interactive app / planned dashboard page.

1.2 Data Source

The dataset is downloaded directly from the **NASA Exoplanet Archive** (Caltech / IPAC), specifically the Planetary Systems (PS) table. This is real observational data, not synthetic — and because NASA confirms new exoplanets on an ongoing basis, the archive itself changes daily. The team downloaded a snapshot of the table and selected the columns most relevant to habitability analysis.

Attribute	Detail
Dataset Name	NASA Exoplanet Archive — Planetary Systems (PS) Table
Source Platform	NASA Exoplanet Science Institute (Caltech / IPAC)
Dataset URL	exoplanetarchive.ipac.caltech.edu/cgi-bin/TblView/nph-tblView?app=ExoTbls&config=PS

Attribute	Detail
Data Type	Real observational data (continuously updated by NASA)
Update Frequency	Updated daily as new exoplanets are confirmed
Snapshot Used	6,128 planets × 19 selected columns (initial raw download)
Secondary Source	Parallax.csv — 37,067 rows (pl_name, sy_plx), used to fill missing system distances
File Format	CSV

Note: Because the NASA archive is a live, daily-updating source, row counts and summary statistics captured at different points in the pipeline (raw download, cleaning notebook, Gold layer, final dashboard) will not match exactly — each reflects the snapshot that existed at the time it was pulled. This documentation reports the actual numbers observed at each stage rather than forcing them to agree.

1.2.1 Raw Columns Selected from the Archive

19 columns were selected from the archive covering identifiers, discovery metadata, orbital properties, planet physical properties, and host star properties:

Column Name	Description
pl_name	Planet name (identifier)
Star's name	Host star name
sy_snum	Number of stars in the system
sy_pnum	Number of planets in the system
discoverymethod	Detection method (Transit, Radial Velocity, Imaging, ...)
disc_year	Year the planet was discovered
pl_orb_period (days)	Orbital period
pl_orb_semi_maj_axis (AU)	Semi-major axis of the orbit
pl_orb_eccent	Orbital eccentricity (0 = circular)
pl_rad (Earth Radius)	Planet radius, Earth units
pl_bmass (Earth Mass)	Planet mass, Earth units
pl_dens	Planet density
st_teff (Kelvin)	Host star effective temperature
st_rad (solar radius)	Host star radius
st_mass (solar mass)	Host star mass
sy_dist (Parsec)	System distance from Earth
pl_teff (Kelvin)	Planet equilibrium temperature (recalculated later)
st_lum (solar luminosity)	Stellar luminosity (recalculated later)
pl_recei_flux	Planet's received stellar flux (recalculated later)

1.3 Project Impact (KPIs)

The KPIs below are the headline numbers surfaced on the dashboard, grouped by the theme of each dashboard page.

Discovery & Overview KPIs

KPI	Definition	Derivation
Total Planets	Total confirmed exoplanets in the dataset	COUNTROWS(Fact_Exoplanet)
Unique Host Stars	Distinct number of host stars	DISTINCTCOUNT(Dim_Star[star_name])
Max Distance (pc)	Furthest system from Earth, in parsecs	MAX(Dim_Star[System_Distance])
Binary System Share	% of star systems with 2+ stars	COUNT(Is_binary = TRUE) / COUNT(all stars)

Physical & Orbital KPIs

KPI	Definition	Derivation
Avg Radius	Mean planet radius (Earth radii)	AVERAGE(Dim_Planet[Radius])
Avg Mass	Mean planet mass (Earth masses)	AVERAGE(Dim_Planet[Mass])
Avg Density	Mean planet density (g/cm ³)	AVERAGE(Dim_Planet[Density])
Avg Eccentricity	Mean orbital eccentricity	AVERAGE(Dim_Planet[Eccentricity])

Stellar Properties KPIs

KPI	Definition	Derivation
Avg Star Temperature	Mean host star effective temperature (K)	AVERAGE(Dim_Star[Star Temperature])
Avg Stellar Mass	Mean host star mass (solar masses)	AVERAGE(Dim_Star[Stellar Mass])
Avg Stellar Luminosity	Mean host star luminosity (solar luminosities)	AVERAGE(Dim_Star[Stellar Luminosity])
Largest System	Highest number of planets discovered around one star	MAX(Dim_Planet[Planet numbers])

Habitability KPIs

KPI	Definition	Derivation
Habitable Planets	Planets whose Flux and Temperature fall inside the habitable zone	COUNT where is_habitable = TRUE
Rocky Planets	Planets with Radius < 1.6 Earth radii	COUNT where is_rocky = "Rocky"
Highest ESI	Highest Earth Similarity Index in the dataset	MAX(Dim_Planet[ESI])
Closest Habitable Planet Distance	Distance (pc) of the nearest habitable-zone planet	MIN(System_Distance) where is_habitable = TRUE

1.4 Main Dashboard Pages Overview

The Power BI dashboard is organized as a data storytelling journey across seven pages — a cover, a mission page, four analytical pages, and a planned AI prediction page:

#	Page Title	Focus Area
Cover	Exoplanet — Cover Page	Project identity and branding
1	Description	Mission statement and project scope
2	Overview	Discovery history, distances, single vs. multi-star systems
3	Habitability (Physical & Orbital)	Planet sizes, masses, density, orbital shape, rocky vs. gaseous
4	Star Systems	Host star temperature, mass, luminosity, correlations
5	Planet Explorer (Habitability & Goldilocks Zone)	Flux vs. temperature, ESI ranking, top Earth-like candidates
6	AI Model	Interactive habitability & similarity prediction for a user-entered planet

Note: Two of the internal notebook/documentation names differ slightly from the final dashboard page names: the notebook's "Page 2: Physical & Orbital Characteristics" became the dashboard's "Habitability" page, and "Page 4: Habitability & Goldilocks Zone" became "Planet Explorer." This documentation uses the final dashboard names.

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Chapter 2

Chapter 2 — Technical Architecture

2.1 Tools Used

The pipeline was built with **Python notebooks (pandas / NumPy)** rather than a managed lakehouse platform — a lighter-weight stack appropriate for a dataset in the thousands-of-rows range.

Tool / Library	Role in Project
NASA Exoplanet Archive	Primary data source — real, daily-updating observational data
Python (pandas, NumPy)	Data cleaning, scientific feature engineering, and the Bronze/Silver/Gold pipeline
Jupyter / Google Colab notebooks	Notebook environment for all data engineering and analysis steps
Matplotlib / Seaborn	EDA charting for the dashboard-design phase
scikit-learn	RandomForestClassifier, StandardScaler, train/test split, evaluation metrics
XGBoost	Gradient-boosted regressor for the Earth Similarity Score model
imbalanced-learn (SMOTE)	Oversampling the rare "habitable" class during classifier training
joblib	Serializing the classifier and scaler to .pkl files
Streamlit	Lightweight interactive web app (app.py) that serves both ML models
Power BI	Final dashboard — DAX measures, star-schema data model, all visuals
Figma	Dashboard layout, color palette, and page design, prototyped before Power BI build

2.2 Architecture (Medallion)

The Medallion Architecture organizes the pipeline into three logical layers — Bronze, Silver, and Gold — implemented as three separate notebooks, each producing its own CSV/Excel output that feeds the next stage.

Layer	Notebook	Purpose	Output
Bronze	Bronze_Layer.ipynb	Raw ingestion: load the cleaned NASA extract and the parallax file, inspect structure and nulls, no transformation logic applied	new_cleaned_exoplanets.csv, new_parallax.csv
Silver	Silver_Layer.ipynb	All imputation logic: physical-relationship fills, Kepler's Third Law, eccentricity assumption, recompute density/luminosity/flux/temperature, drop unrecoverable rows, merge parallax, remove duplicates	Data_Cleaned_Silver_layer.csv
Gold	Gold_Layer.ipynb	Dimensional modeling: split into star-schema fact + dimension tables	Data_Gold_Layer.csv / .xlsx

A fourth step — not part of the strict medallion naming but functionally an enrichment pass on top of Gold — runs inside the EDA notebook and adds the habitability science layer (ESI, escape velocity, habitability/rocky flags) described in Chapter 3.

2.3 Data Model (Star Schema)

The Gold layer implements a star schema: one central Fact table surrounded by four Dimension tables, all linked through surrogate keys generated during the Gold_Layer notebook. This mirrors the star-schema table names actually found inside the delivered Power BI file (Exoplanet_Dashboard_final.pbix).

Table	Type	Key Columns	Description
Fact_Exoplanets	Fact	pl_name, planet_id (FK), star_id (FK), discovery_id (FK), Date_id (FK)	Central table linking every planet to its star, discovery record, and discovery date
Dim_Planet	Dimension	planet_id, PlanetName	Planet numbers, orbital period, semi-major axis, eccentricity, radius, mass, density, temperature, flux — plus the enriched habitability columns (ESI, is_habitable, is_rocky, is_habitable_rocky, Escape_Velocity)
Dim_Star	Dimension	star_id, Star_Name	Star temperature, radius, mass, luminosity, system distance, parallax, number of stars, Is_binary flag
Dim_Discovery	Dimension	discovery_id	Discovery method and discovery year
Dim_Date	Dimension	Date_id	Discovery year, used for the time-series / cumulative-discoveries visuals

Note: The Gold_Layer notebook builds the dimension tables first (with drop_duplicates on their natural keys), then constructs the fact table by looking up each dimension's surrogate key

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Chapter 3

Chapter 3 — Data Engineering & Analysis

Data Cleaning Process

GitHub: [Copy_of_Data_Cleaning.ipynb](#) (initial exploration) → mirrored inside [Silver_Layer.ipynb](#) for the pipeline

The raw snapshot downloaded from the NASA Exoplanet Archive contained 6,128 planets and 19 columns. Initial exploration showed extensive missingness across the orbital and physical columns — the team's own estimate was that roughly 43% of cells across the dataset were missing or held placeholder zeros. Rather than dropping incomplete rows outright, the team solved each gap using the underlying physics wherever possible, and only dropped rows as a last resort.

Column	Missing (this snapshot)	% Missing
pl_orb_semi_maj_axis (AU)	2,307 / 6,128	37.6%
pl_orb_eccent	3,542 / 6,128	57.8%
pl_rad (Earth Radius)	1,575 / 6,128	25.7%
pl_bmass (Earth Mass)	3,102 / 6,128	50.6%
st_teff (Kelvin)	711 / 6,128	11.6%
st_rad (solar radius)	798 / 6,128	13.0%
st_mass (solar mass)	773 / 6,128	12.6%
sy_dist (Parsec)	135 / 6,128	2.2%

Note: These exact percentages come from the specific archive snapshot captured in the uploaded notebook. Because NASA updates the archive daily, a fresh download will show slightly different counts — the imputation logic below is what stays constant.

Step A — Replace placeholder zeros with NaN

Several columns used 0 as a placeholder for "unmeasured" rather than a true zero value, so these were converted to proper nulls before any statistics were computed.

```
df['pl_orb_period(in E.years)'] = pd.to_numeric(df['pl_orb_period(in E.years)'], errors='coerce')
df['pl_orb_period(in E.years)'] = df['pl_orb_period(in E.years)'].replace(0, np.nan)
df.replace(0, np.nan, inplace=True)
```

Step B — Stellar radius & mass (Radius–Mass relation for main-sequence stars)

~12–13% of stellar radius/mass values were missing. For main-sequence stars, radius and mass follow a well-known power-law relationship, used here in both directions to cross-fill whichever value is missing.

```
df['st_mass(in solar mass)'] = df['st_mass(in solar mass)'].fillna(df['st_rad(in solar radius)']**(1/0.8))
df['st_rad(in solar radius)'] = df['st_rad(in solar radius)'].fillna(df['st_mass(in solar mass)']**0.8)
```

Step C — Semi-major axis & orbital period (Kepler's Third Law)

~38–40% of orbit-size and orbital-period values were missing. Kepler's Third Law directly relates a planet's orbital period, its semi-major axis, and its star's mass — so each missing value can be recovered from the other two, applied iteratively so that both columns benefit.

```
calc_a = (df['st_mass(in solar mass)'] * (df['pl_orb_period(in days)'] / 365.25)**2)**(1/3)
df['pl_orb_semi_maj_axis(in AU)'] = df['pl_orb_semi_maj_axis(in AU)'].fillna(calc_a)

calc_p = 365.25 * np.sqrt((df['pl_orb_semi_maj_axis(in AU)']**3) / df['st_mass(in solar mass)'])
df['pl_orb_period(in days)'] = df['pl_orb_period(in days)'].fillna(calc_p)
df['pl_orb_period(in E.years)'] = df['pl_orb_period(in days)'] / 365.25
```

Step D — Eccentricity (near-circular assumption for small orbits)

~58% of eccentricity values were missing in this snapshot. Earth's own orbit is nearly circular (eccentricity ≈ 0.017), and planets with small orbits (semi-major axis < 1.15 AU) are physically unlikely to have highly elliptical orbits. Missing values for these close-in planets were filled with a small random value in $[0, 0.1]$ rather than a single constant, to avoid an artificial spike at exactly zero.

```
cond = df['pl_orb_eccent'].isnull() & (df['pl_orb_semi_maj_axis(in AU)'] < 1.15)
num_to_fill = cond.sum()
random_ecc = np.random.uniform(0.0, 0.1, size=num_to_fill)
df.loc[cond, 'pl_orb_eccent'] = random_ecc
```

Step E — Planet radius & mass (Mass–Radius relation, split by regime)

~26–51% of planet radius/mass values were missing. Small, rocky planets and large, gaseous planets follow different mass–radius power laws, so the fill uses a 2-Earth-radius (and ~13-Earth-mass) split point to pick the right relationship in each direction.

```
# Rocky planets where R < 2
condition_rocky_R = (df['pl_rad(in Earth Radius)'] < 2) & df['pl_bmass(in Earth Mass)'].isnull()
df.loc[condition_rocky_R, 'pl_bmass(in Earth Mass)'] = df.loc[condition_rocky_R, 'pl_rad(in Earth Radius)']**3.68

# Gas giants where R >= 2
condition_gas_R = (df['pl_rad(in Earth Radius)'] >= 2) & df['pl_bmass(in Earth Mass)'].isnull()
df.loc[condition_gas_R, 'pl_bmass(in Earth Mass)'] = 2.69 * (df.loc[condition_gas_R, 'pl_rad(in Earth Radius)']**0.93)

# Same logic run in reverse to fill missing radius from known mass
condition_rocky_M = (df['pl_bmass(in Earth Mass)'] < 13) & df['pl_rad(in Earth Radius)'].isnull()
df.loc[condition_rocky_M, 'pl_rad(in Earth Radius)'] = df.loc[condition_rocky_M, 'pl_bmass(in Earth Mass)']**(1/3.68)

condition_gas_M = (df['pl_bmass(in Earth Mass)'] >= 13) & df['pl_rad(in Earth Radius)'].isnull()
df.loc[condition_gas_M, 'pl_rad(in Earth Radius)'] = (df.loc[condition_gas_M, 'pl_bmass(in Earth Mass)'] / 2.69)**(1/0.93)
```

Step F — Drop remaining unrecoverable rows

After exhausting the physics-based fills, a small number of rows still had no usable stellar or planetary measurements at all. These rows carry no analytical value and were dropped rather than left partially null.

```
df = df.dropna(subset=['st_mass(in solar mass)', 'st_rad(in solar radius)', 'st_teff(in kelvin)'])
df = df.dropna(subset=['pl_bmass(in Earth Mass)', 'pl_rad(in Earth Radius)'], how='all')
```

This snapshot's cleaning pass reduced the dataset from 6,128 rows to 5,394 rows with zero remaining nulls in the core physical columns.

New Columns Derived During Cleaning

Three physical quantities are not directly measured by NASA but can be derived from what is measured, and were added at this stage:

New Column	Law Used	Formula
Stellar Luminosity	Stefan–Boltzmann Law	$L = R_{\star}^2 \times (T_{\star} / 5778)^4$ (solar units)
Planet Received Flux	Inverse-Square Law	$\text{Flux} = L_{\star} / a^2$ (a = semi-major axis, AU)
Planet Equilibrium Temperature	Stellar-flux balance	$T_{\text{eq}} = T_{\star} \times \sqrt{(R_{\star} \times 0.00465) / (2a)}$

```
df['pl_dens(in (Earth mass/Earth radius))'] = 5.51 * (df['pl_bmass(in Earth Mass)'] / (df['pl_rad(in Earth Radius)']**3))
df['st_lum(in Solar luminosity)'] = (df['st_rad(in solar radius)']**2) * ((df['st_teff(in kelvin)'] / 5778)**4)
df['pl_recei_flux(in Earth\'s received flux)'] = df['st_lum(in Solar luminosity)'] / (df['pl_orb_semi_maj_axis(in AU)']**2)
df['pl_teff(in kelvin)'] = df['st_teff(in kelvin)'] * np.sqrt((df['st_rad(in solar radius)'] * 0.00465) / (2 * df['pl_orb_semi_maj_axis(in AU)']))
```

Merging in System Distance (Parallax → Distance-Modulus Law)

~2–3% of system distances were still missing. These were recovered by merging in a second NASA export, Parallax.csv (37,067 rows, pl_name + parallax in milliarcseconds), and applying the parallax-to-distance relationship.

```
df_merged = pd.merge(df, df_new[['pl_name', 'sy_plx']], on='pl_name', how='left')
df_merged['sy_plx'] = pd.to_numeric(df_merged['sy_plx'], errors='coerce')
valid_parallax = df_merged['sy_plx'] > 0
distance = 1000 / df_merged.loc[valid_parallax, 'sy_plx']
df_merged['sy_dist(in Parsec)'] = df_merged['sy_dist(in Parsec)'].fillna(distance)
```

Note: Parallax.csv contains multiple parallax measurements per planet name (37,067 rows for far fewer unique planets), so the left-merge temporarily multiplies row count before duplicates are removed in the next step.

Removing Duplicates

The merge above produces duplicate rows wherever a planet had more than one parallax record. These were removed with a final deduplication pass before saving the cleaned file for the Medallion pipeline.

```
d.drop_duplicates(inplace=True)
```

The final cleaned file — Exoplanet_data_new_cleaned.csv — is the file the Bronze layer ingests to begin the formal Medallion pipeline described in Chapter 2.

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Chapter 4

Chapter 4 — Feature Engineering — Habitability Science Layer

GitHub: [EDA Habitability Analysis.ipynb](#) | Input: [Data Gold Layer.csv](#) (5,878 planets) | Output: [Data Gold Layer Enriched.csv](#)

On top of the Gold layer, the EDA notebook fills any remaining gaps (Temperature, Flux, Eccentricity → 0, System_Distance, Parallax with medians) and adds the columns that turn raw planet measurements into a habitability verdict.

Escape Velocity

```
df['Escape_Velocity'] = np.sqrt(df['Mass'] / df['Radius'])
```

Earth Similarity Index (ESI)

The ESI is a 0–1 score (1.0 = identical to Earth) that combines four planet properties, each compared to Earth's own value and weighted by how strongly it affects habitability. Temperature carries by far the largest weight because it matters most for liquid water and life as we know it.

Property	Earth Reference Value	Weight
Radius	1.0 Earth radii	0.57
Density	5.51 g/cm ³	1.07
Escape Velocity	1.0 (Earth = 1)	0.70
Temperature	255 K	5.58 (dominant factor)

```
R_ref, D_ref, V_ref, T_ref = 1.0, 5.51, 1.0, 255.0
w_R, w_D, w_V, w_T = 0.57, 1.07, 0.70, 5.58
w_sum = w_R + w_D + w_V + w_T

esi_R = (1 - np.abs((df['Radius'] - R_ref) / (df['Radius'] + R_ref))) ** w_R
esi_D = (1 - np.abs((df['Density'] - D_ref) / (df['Density'] + D_ref))) ** w_D
esi_V = (1 - np.abs((df['Escape_Velocity'] - V_ref) / (df['Escape_Velocity'] + V_ref))) ** w_V
esi_T = (1 - np.abs((df['Temperature'] - T_ref) / (df['Temperature'] + T_ref))) ** w_T

df['ESI'] = (esi_R * esi_D * esi_V * esi_T) ** (1 / w_sum)
```

Habitability Flags

```
df['is_habitable'] = (df['Flux'] >= 0.25) & (df['Flux'] <= 2.2) & (df['Temperature'] >= 180) &
(df['Temperature'] <= 320)
df['is_rocky'] = df['Radius'] < 1.6
df['is_habitable_rocky'] = df['is_habitable'] & df['is_rocky']
```

Flag	Rule	Result in EDA run (5,878 planets)
is_habitable	$0.25 \leq \text{Flux} \leq 2.2$ AND $180\text{K} \leq \text{Temperature} \leq 320\text{K}$	302 planets
is_rocky	Radius < 1.6 Earth radii	separates rocky planets from gas giants
is_habitable_rocky	is_habitable AND is_rocky	45 planets

The single highest-ESI planet in the dataset is GJ 1002 b, with ESI = 0.9935 — essentially Earth's closest known physical twin among confirmed exoplanets in this dataset. The closest confirmed habitable-zone planet is Proxima Centauri b, at roughly 1.30 parsecs (about 4.2 light-years).

Note: The habitability and ESI logic live directly in the source data columns (is_habitable, is_rocky) loaded into Power BI via Power Query — not as DAX calculated columns. Inspecting the final .pbix confirmed the Power Query step correctly maps is_rocky (true/false) to the "Rocky" / "Gaseous" labels shown on the dashboard, with no inverted logic.

Chapter 5

Chapter 5 — Machine Learning

5.1 Models Overview

The AI component was scoped as two complementary models, matching the team's original concept design:

Model	Question it Answers	Input	Output
1 — Habitability Classifier	Is this planet Habitable or Not?	Radius, Mass, Flux, Temperature, Semi-major Axis, Eccentricity	Habitable ☒ / Not Habitable ☒ + Confidence %
2 — Earth Similarity Score (Regression)	How close to Earth-like is it, on a 0–100 scale?	Planet & star properties (same 6 inputs)	Earth Similarity Score, e.g. 87.5 / 100

The concept stage considered Random Forest / XGBoost for the classifier and Random Forest Regressor / Gradient Boosting for the regressor. The implementation notebook (AiModel.ipynb) settled on:

- Habitability Classifier → RandomForestClassifier (scikit-learn), with SMOTE oversampling to counter class imbalance.
- Earth Similarity Regressor → XGBRegressor (XGBoost).

5.2 Data & Preprocessing

The models are trained on ExoHabX — Habitable Exoplanets Dataset (NASA / HWC), downloaded from Kaggle via kagglehub (kaggle.py): 5,326 planets across 9 columns (pl_name, pl_rade, pl_bmasse, pl_eqt, pl_orbsmax, pl_orbeccen, st_teff, sy_snum, P_ESI). P_ESI is a pre-computed Earth Similarity Index sourced from the Habitable Worlds Catalog.

```
df_raw = pd.read_csv('exoplanet_dataset.csv')

df = pd.DataFrame()
df['radius'] = df_raw['pl_rade']
df['mass'] = df_raw['pl_bmasse']
df['flux'] = df_raw['st_teff']
df['temp'] = df_raw['pl_eqt']
df['semi_major'] = df_raw['pl_orbsmax']
df['eccentricity'] = df_raw['pl_orbeccen']

df['ESI_Score'] = df_raw['P_ESI'] * 100
df['Habitability_Label'] = np.where(df['ESI_Score'] >= 80, 1, 0)
df = df.dropna()
```

Result: 25 of the 5,326 planets (0.47%) are labeled habitable under this $ESI \geq 80$ rule — an even more imbalanced target than before, addressed with a stratified train/test split and SMOTE oversampling:

```
X = df.drop(['Habitability_Label', 'ESI_Score'], axis=1)
y_class = df['Habitability_Label']
y_reg = df['ESI_Score']

X_train, X_test, y_class_train, y_class_test, y_reg_train, y_reg_test = train_test_split(
    X, y_class, y_reg, test_size=0.2, stratify=y_class, random_state=42)

scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
```

```
k_neighbors = min(5, habitable_count - 1)
smote = SMOTE(random_state=42, k_neighbors=k_neighbors)
X_train_class_resampled, y_class_train_resampled = smote.fit_resample(X_train_scaled, y_class_train)
```

Original habitable count in the training split: 20 planets. After SMOTE: 4,240 (balanced 50/50 with the majority class).

5.3 Model Results

Habitability Classifier (Random Forest, regularized)

The classifier was tuned with explicit regularization and a custom decision threshold to check for and reduce overfitting:

```
clf_model = RandomForestClassifier(
    n_estimators=150,
    class_weight='balanced',
    max_depth=7,
    min_samples_split=10,
    random_state=42
)
clf_model.fit(X_train_class_resampled, y_class_train_resampled)

probs = clf_model.predict_proba(X_test_scaled)[: , 1]
custom_threshold = 0.4
test_preds = (probs >= custom_threshold).astype(int)
```

Check	Value		
Training Accuracy	99.88%		
Testing Accuracy	99.06%		
Confusion Matrix	[[1053, 8], [2, 3]] (rows = actual, columns = predicted)		
Metric	Class 0 (Not Habitable)	Class 1 (Habitable)	Overall
Precision	1.00	0.27	—
Recall	0.99	0.60	—
F1-score	1.00	0.38	—
Support	1,061	5	1,066
Accuracy	—	—	0.99
Macro Avg (P / R / F1)	—	—	0.64 / 0.80 / 0.69

Earth Similarity Regressor (XGBoost, sample-weighted)

The regressor was re-tuned with sample weights that penalize errors on high-ESI (highly Earth-like) planets 20× more heavily than on typical planets, so the model is pushed to pay closer attention to the rare, high-value cases:

```
weights = np.where(y_reg_train >= 70, 20, 1)

reg_model = XGBRegressor(
    n_estimators=500,
    learning_rate=0.05,
    max_depth=8,
```

```

subsample=0.9,
colsample_bytree=0.9,
random_state=42
)
reg_model.fit(X_train_scaled, y_reg_train, sample_weight=weights)

```

Metric	Value
RMSE	4.58 (on the 0–100 ESI_Score scale)
MAE	2.31
R ² Score	87.00%

Sanity-Check Predictions

Before saving, the notebook runs the pipeline end-to-end on two held-out test planets to confirm the models behave sensibly on real, unseen cases:

Test Case	True Label	True ESI	Predicted Label	Predicted ESI
1 — Habitable planet	Habitable (1)	96.84 / 100	Habitable (99.6% confidence)	89.25 / 100
2 — Not-habitable planet	Not Habitable (0)	14.05 / 100	Not Habitable (100.0% confidence)	17.35 / 100

5.4 Why These Algorithms

- 5.4.1 Random Forest Classifier — robust to a small, noisy, imbalanced feature set; requires little tuning; handles the SMOTE-resampled training data well; matches the concept stage's primary recommendation.
- 5.4.2 XGBoost Regressor — strong baseline for tabular regression with non-linear feature interactions; consistent with the concept stage's Gradient Boosting recommendation; sample weighting lets it prioritize accuracy on the rare high-ESI planets that matter most for the project's mission.

5.5 Model Artifacts & Deployment

The trained models and supporting scaler were serialized for reuse outside the training notebook:

File	Contents
habitability_classifier.pkl	Trained RandomForestClassifier (joblib)
feature_scaler.pkl	Fitted StandardScaler used for both models
earth_similarity_regressor.json	Trained XGBRegressor, saved via XGBoost's native format

```

joblib.dump(clf_model, 'habitability_classifier.pkl')
joblib.dump(scaler, 'feature_scaler.pkl')
reg_model.save_model('earth_similarity_regressor.json')

```

Streamlit Demo App ([app.py](#))

A small Streamlit app loads all three artifacts and exposes an interactive form: the user enters Planet Radius, Mass, Stellar Flux, Equilibrium Temperature, Semi-major Axis, and Eccentricity, and the app returns the classifier's Habitable

/ Not Habitable verdict with confidence %, plus the regression model's Earth Similarity Index (0–100) with a progress bar.

```
clf = joblib.load('habitability_classifier.pkl')
scaler = joblib.load('feature_scaler.pkl')
reg = xgb.XGBRegressor()
reg.load_model('earth_similarity_regressor.json')

input_scaled = scaler.transform(input_data)
is_habitable = clf_model.predict(input_scaled)[0]
confidence = clf_model.predict_proba(input_scaled)[0].max() * 100
esi_score = reg_model.predict(input_scaled)[0]
```

This same six-input design is now mirrored directly inside the Power BI report itself — the "AI Model" page (see Chapter 5, section 5.4.7) runs the identical classifier and regressor through a Python visual, using six live slicers as inputs.

Chapter 6 — Dashboard & Visualization

The dashboard is the end-user deliverable of the project — a seven-page interactive Power BI report (Exoplanet_Dashboard_final.pbix) that turns the Gold-layer, habitability-enriched dataset into a guided data story. This chapter documents the design philosophy, the confirmed Power BI structure (pages, visuals, and DAX measures extracted directly from the .pbix file), and a page-by-page walkthrough with screenshots.

6.1 Power BI Dashboard Pages

The [dashboard](#) connects directly to the Gold-layer star schema described in Chapter 2. Inspecting the .pbix file confirms seven report pages, in this order:

Page	Title	Status
Cover	Exoplanet	Complete — branding & mission page
1	Description	Complete — mission statement
2	Overview	Complete — 4 KPI cards + 4 charts + 4 slicers
3	Habitability	Complete — 4 KPI cards + 4 charts + 4 slicers
4	Star Systems	Complete — 4 KPI cards + 3 charts + 1 Python visual + 4 slicers
5	Planet Explorer	Complete — 4 KPI cards + 4 charts + 4 slicers
6	AI Model	Complete — 6 live slicers + 1 Python visual running the trained classifier and regressor

6.2 KPI Highlights (as shown on the live dashboard)

KPI	Value	Dashboard Page
Total Planets	5,877	Overview
Unique Host Stars	4,024	Overview
Max Distance (pc)	7,200	Overview
Binary System Share	8.55%	Overview
Avg. Radius	97.93 Earth radii	Habitability
Avg. Mass	251.57 Earth masses	Habitability
Avg. Density	5.38 g/cm ³	Habitability
Avg. Eccentricity	0.08	Habitability
Avg. Star Temperature	5,402.75 K	Star Systems
Avg. Stellar Mass	0.99 solar masses	Star Systems
Avg. Stellar Luminosity	8.12 solar luminosities	Star Systems
Largest System	8 planets around one star	Star Systems
Habitable Planets	281	Planet Explorer
Rocky Planets	43	Planet Explorer

KPI	Value	Dashboard Page
Highest ESI	0.99	Planet Explorer
Closest Habitable Planet	≈ 1 pc (card display, rounded)	Planet Explorer

6.3 Figma Design

Figma file: ExoPlanet Dashboard — figma.com/design/XFcLLV4rbpQi2xB6Obne5g/ExoPlanet-Dashboard

The full dashboard was designed and prototyped in Figma before any visual was placed in Power BI, establishing the storytelling flow, color palette, and page layout up front.

Visual Identity

- **Cosmic / nebula theme** — a deep space-black background with a soft purple-and-violet nebula wash and scattered stars runs across every page, evoking the astronomical subject matter.
- **Color palette** — indigo and violet purples for KPI cards and chart series, with a warm gold/amber accent reserved for callouts and the project's supervisor/team branding.
- **Consistent left sidebar** — every analytical page repeats the same set of thin line-art icons (planet with rings, telescope, magnifying globe, DNA helix, rocket, star, funnel/filter) as a fixed navigation rail, giving the whole report a unified identity.
- **Orbit / atom motif** — the cover and description pages both use a circular "orbit" diagram (icons arranged in rings around a central planet or star) to visually represent the project's six analytical dimensions before the user reaches the data.
- **Tagline** — "Where Data Meets Discovery" appears in cursive script on the cover page as the project's mission tagline.

6.4 Dashboard Walkthrough

6.4.1 Cover Page — "Where Data Meets Discovery"

The cover page establishes the project's identity: the title "EXOPLANET-HABITABILITY ZONE" in a glowing outlined box, the eYouth and Digital Egypt Pioneers / Ministry of Communications logos, the cursive tagline, the six project team member names, and the supervisor's name. A ringed-planet-and-orbit icon sits beside a ring of circular icons connected by dotted orbit lines, each capped with a LinkedIn badge — a decorative nod to the team's social presence. No KPIs or charts appear on this page; it is a pure branding entry point.

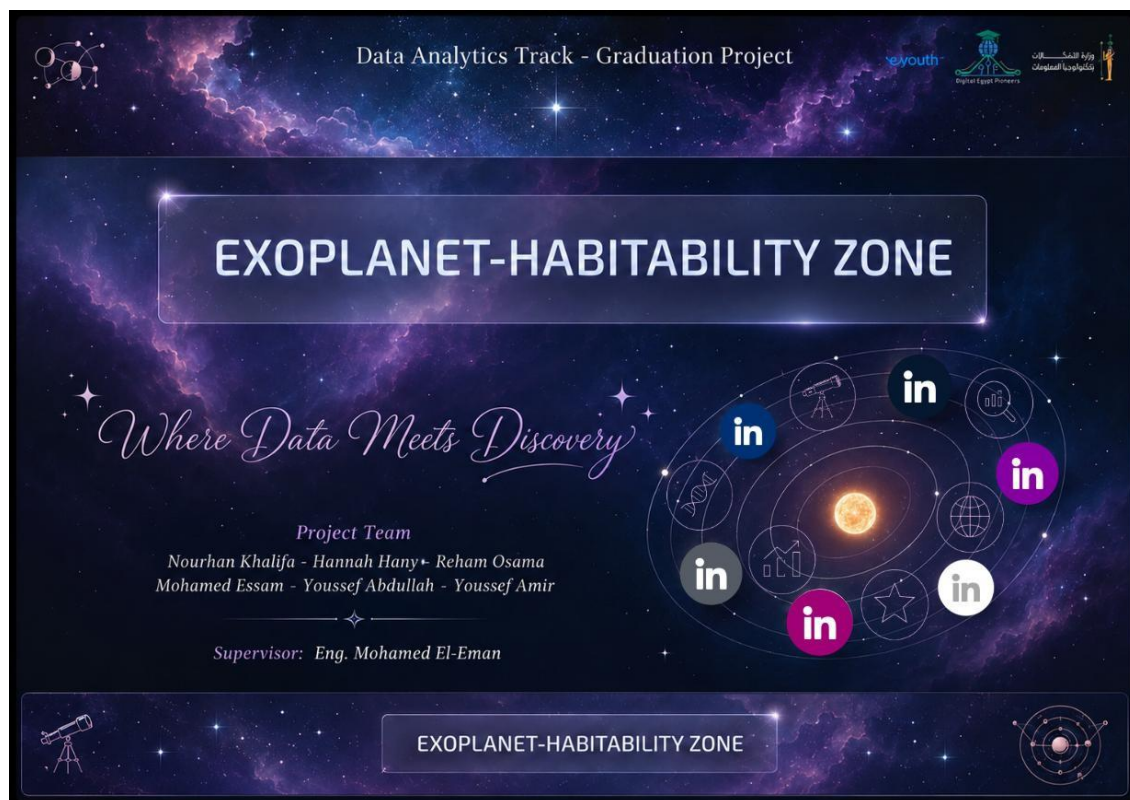


Figure 5.1 — Cover Page: project title, tagline, team & supervisor names, orbit-icon motif

6.4.2 Page 1 — Description: Mission Statement

The Description page states the project's mission in plain language, framed inside a bordered text box, next to the same orbit-icon diagram used on the cover:

“Our mission is to explore distant worlds through data. By analyzing exoplanet and stellar properties, we identify planets located within the habitable zone and evaluate their potential to support life. Turning astronomical data into discoveries, we reveal the conditions that make planets more Earth-like and potentially habitable.”

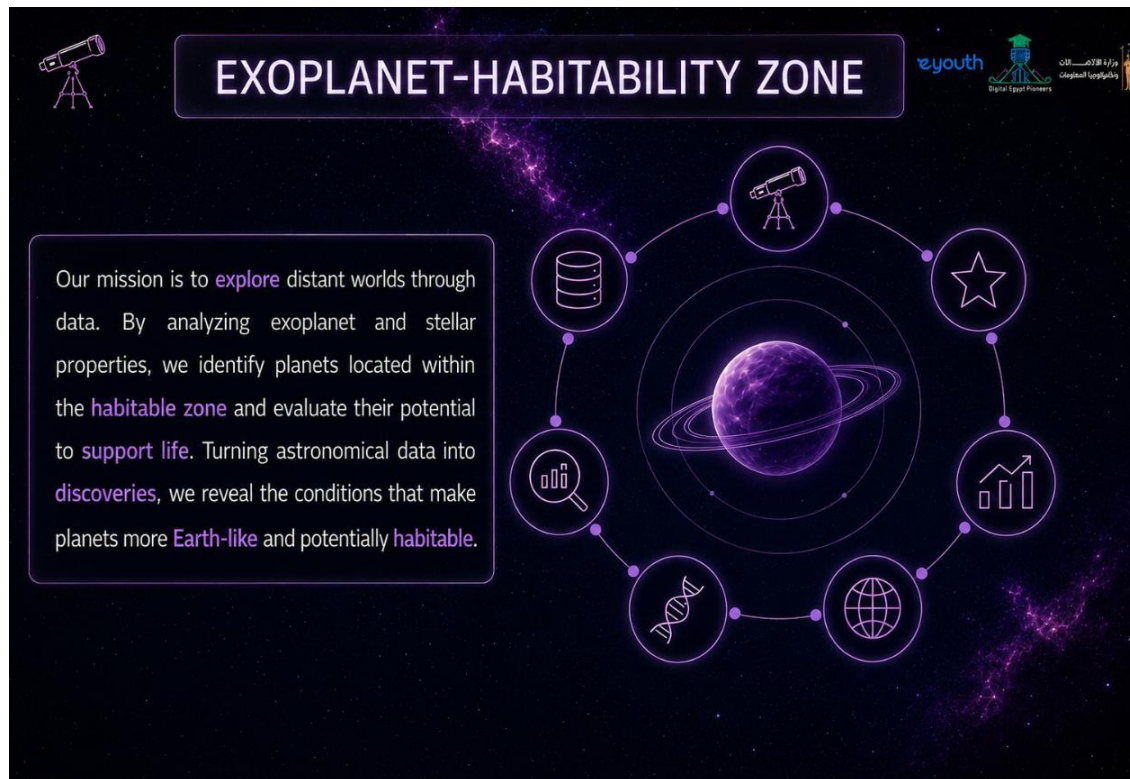


Figure 5.2 — Description Page: mission statement and thematic orbit-icon diagram

6.4.3 Page 2 — Overview: Discovery History & System Structure

The Overview page gives a high-level snapshot of the whole dataset — how many planets and stars exist in it, how far away they are, and how discoveries have accumulated over time.

Element	Type	Key Insight
Total Planets	KPI Card	5,877 confirmed exoplanets in the dataset
Unique Host Stars	KPI Card	4,024 distinct host stars
Max Distance (pc)	KPI Card	7,200 parsecs — the most distant system in the dataset
Binary System Share	KPI Card	8.55% of systems have 2 or more stars
Cumulative Discoveries Over Time by Method	Stacked / Line Area Chart	Discovery pace spikes sharply around 2014–2016 (peak year: 1,522 new planets), driven mainly by the Transit method
Proportion of Single vs Multi-Star Systems	Donut Chart	91.76% of systems have a single star; only 7.18% have two
Success of Discovery Methods	Bar Chart	Transit dominates with ~4.78K planets found, vs. ~1.00K for Radial Velocity and negligible counts for all other methods
Distribution of Exoplanet Distances from Earth	Histogram	Most confirmed planets sit within the first ~1,000 pc; the distribution has a long right tail out to thousands of parsecs
Discovery Year / Method / Num of Stars / System Distance	Slicers	Interactive filtering across all four dimensions

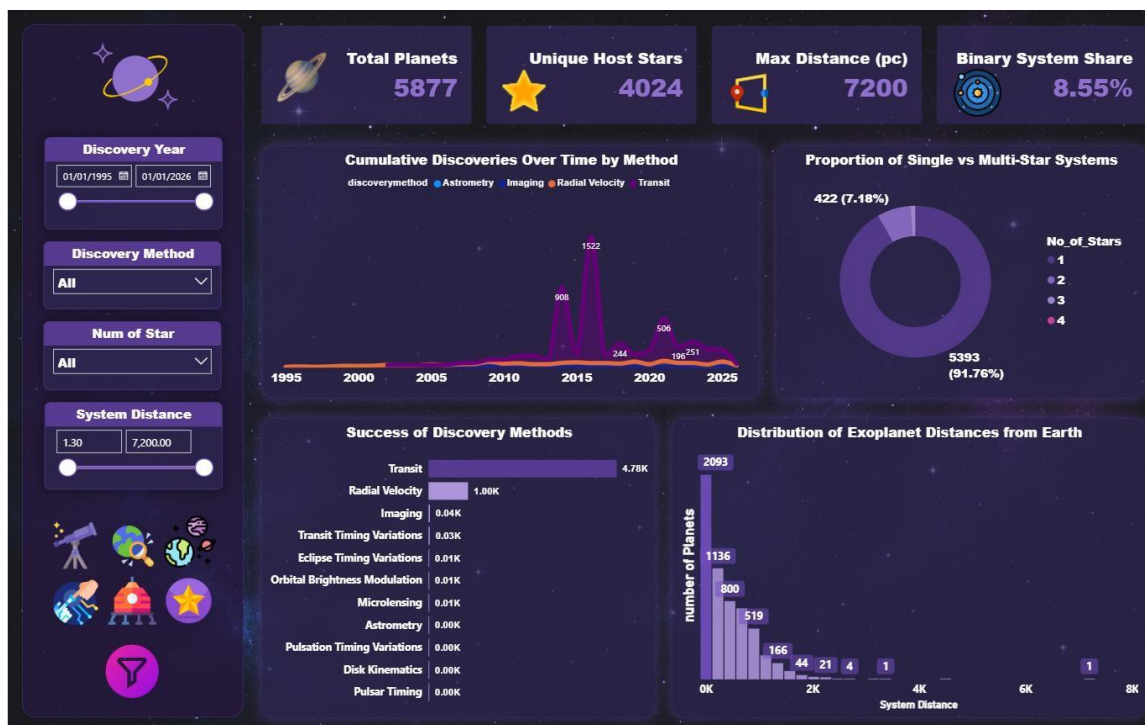


Figure 5.3 — Overview Page: discovery-history KPIs, cumulative-discoveries chart, star-system donut, distance histogram

6.4.4 Page 3 — Habitability (Physical & Orbital Characteristics)

This page moves from "how many" to "what kind" — examining planet size, mass, density, and orbital shape, and validating Kepler's Third Law against the real data.

Element	Type	Key Insight
Avg. Radius	KPI Card	97.93 Earth radii on average — skewed high by the large number of gas giants in the confirmed-planet catalogue
Avg. Mass	KPI Card	251.57 Earth masses on average
Avg. Density	KPI Card	5.38 g/cm ³ — close to Earth's own 5.51 g/cm ³
Avg. Eccentricity	KPI Card	0.08 — most orbits are close to circular
Kepler's Third Law	Scatter Plot	Orbital period rises cleanly with semi-major axis, visually confirming the physical law used to fill missing data in Chapter 3; habitable-zone planets (is_habitable = True) cluster in a distinct band
Mass vs Radius	Scatter Plot	Rocky and gaseous planets separate into two visually distinct populations, colored by is_rocky
Eccentricity Distribution	Histogram	Strongly right-skewed: ~2.8K planets have eccentricity ≈ 0 (near-circular), tapering off quickly toward more elliptical orbits
Rocky vs Gaseous Planets	Bar Chart	4.2K planets classified Gaseous vs. 1.2K Rocky
Radius / Mass / Planet Type / Habitable	Slicers	Interactive filtering by physical properties and habitability flag

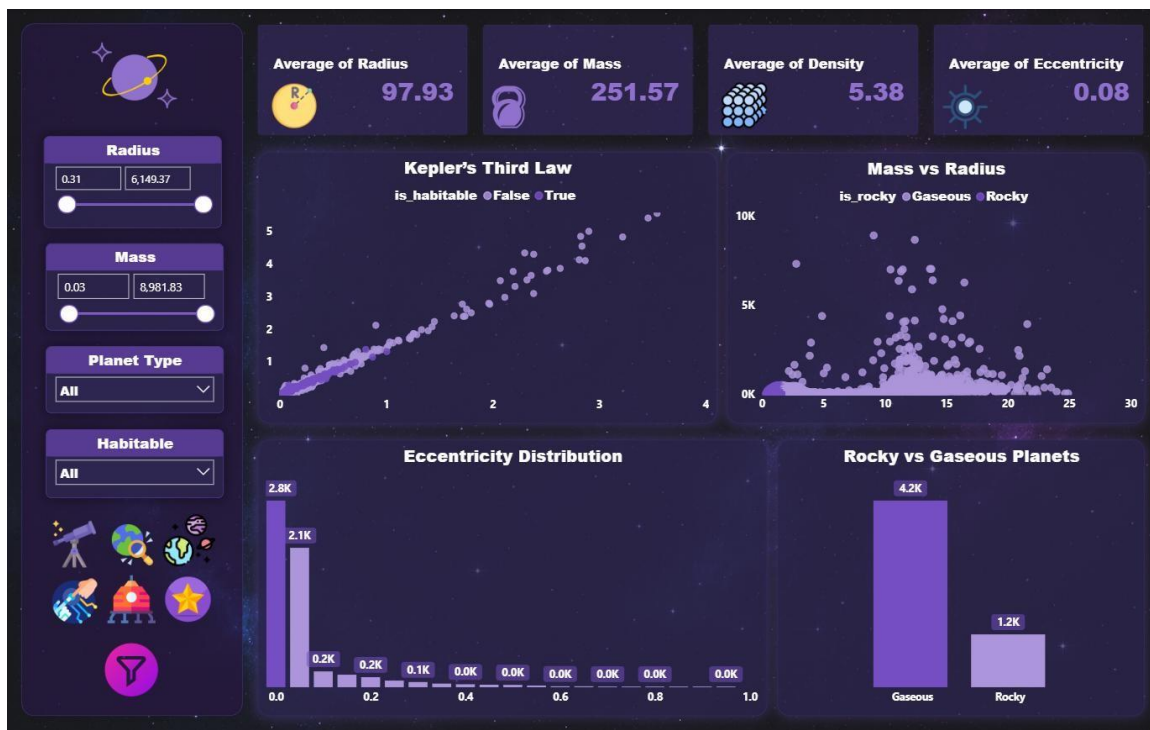


Figure 5.4 — Habitability Page: physical & orbital KPIs, Kepler's Third Law scatter, Mass vs Radius, eccentricity histogram, Rocky vs Gaseous bar

6.4.5 Page 4 — Star Systems: Host Star Properties

This page shifts focus from the planets to the stars they orbit — temperature, mass, luminosity, and how those properties correlate with each other and with system distance.

Element	Type	Key Insight
Average of Star Temperature	KPI Card	5,402.75 K — close to our Sun's 5,778 K
Average of Stellar Mass	KPI Card	0.99 solar masses — almost exactly one Sun
Average of Stellar Luminosity	KPI Card	8.12 solar luminosities
Largest System	KPI Card	8 confirmed planets orbiting a single host star
Host Stars Diagram (Luminosity vs. Temperature)	Scatter Plot	A Hertzsprung–Russell-style diagram; habitable-zone planets (is_habitable) cluster around Sun-like temperatures and moderate luminosity
Stellar Mass vs Temperature	Scatter Plot	Positive relationship — more massive stars tend to run hotter, as expected from stellar physics
Number of Planets per Host Star	Bar Chart	The overwhelming majority of host stars ($\approx 3.1\text{K}$) have only 1 confirmed planet; counts fall off sharply for 2, 3, 4+ planet systems
Rocky vs Gaseous Planets (Correlation Heatmap)	Python Visual	A correlation heatmap across Star Temperature, Stellar Mass, Stellar Luminosity, and System Distance — Stellar Mass and Luminosity show the strongest positive correlation (0.44)
Temperature / Mass / System Distance / Planet Type	Slicers	Interactive filtering across stellar and planetary properties

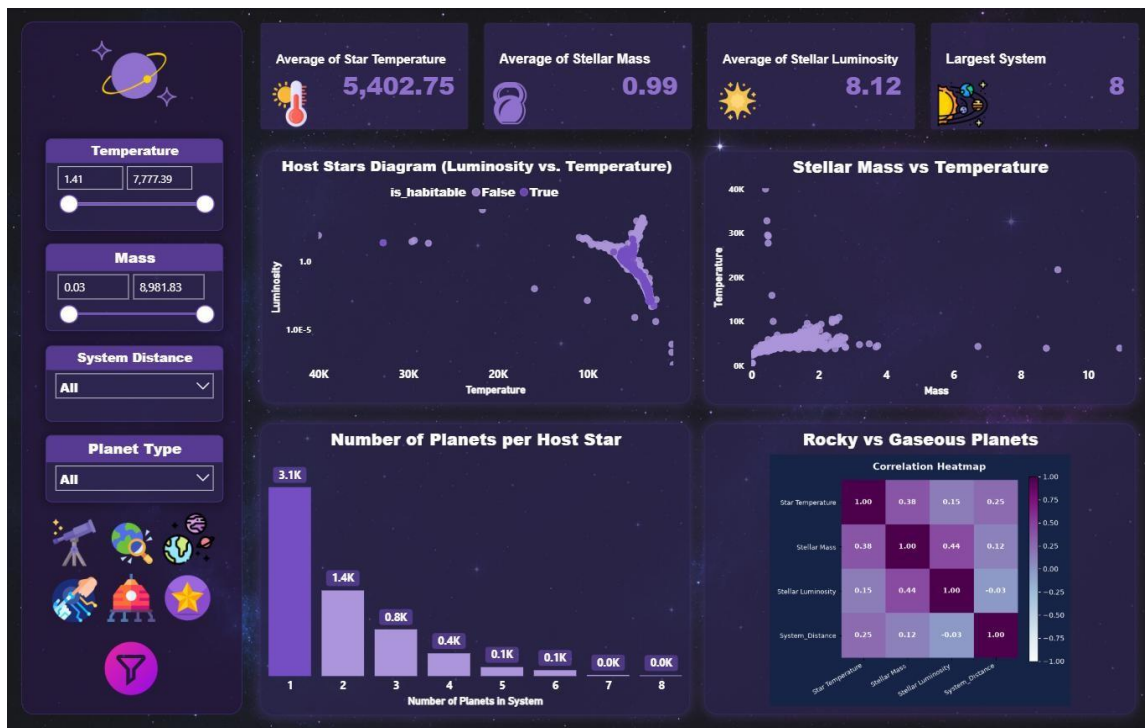


Figure 5.5 — Star Systems Page: stellar KPIs, H–R-style diagram, Stellar Mass vs Temperature, Planets-per-Star bar, correlation heatmap

6.4.6 Page 5 — Planet Explorer (Habitability & Goldilocks Zone)

The analytical centerpiece of the dashboard: this page isolates the planets that could plausibly support life, and ranks them by how Earth-like they are.

Element	Type	Key Insight
Habitable Planets	KPI Card	281 planets fall inside the habitable-zone flux/temperature window
Rocky Planets	KPI Card	43 planets are both habitable and rocky (Radius < 1.6 R _⊕)
Highest ESI	KPI Card	0.99 — the single most Earth-like planet in the dataset
Planet Distance (Closest Habitable)	KPI Card	The nearest confirmed habitable-zone planet, in parsecs
Flux vs Temperature	Scatter Plot	The "Goldilocks Zone" itself — habitable-zone planets (is_habitable = True) trace a tight curve at low flux / moderate temperature
Radius vs Flux	Scatter Plot	Cross-references planet size against stellar flux, colored by is_rocky, to spot small rocky worlds inside the habitable flux range
Earth Similarity Index Distribution	Histogram	Most planets score low on ESI; scores peak around 0.4–0.5 (662 planets), with very few above 0.8
Top 10 Earth-like Planets	Bar Chart	GJ 1002 b (0.99), Wolf 1069 b (0.98), Kepler-1649 c (0.98), TRAPPIST-1 e (0.98), Proxima Cen b (0.97), Teegarden's Star c (0.97), TOI-700 d (0.97), GJ 1061 d (0.96), Kepler-442 b (0.95), GJ 667 C f (0.95)
Temperature / ESI / Habitable / System Distance	Slicers	Interactive filtering by habitability-relevant properties



Figure 5.6 — Planet Explorer Page: habitability KPIs, Goldilocks-Zone scatter, Radius vs Flux, ESI distribution, Top-10 Earth-like planets

6.4.7 Page 6 — AI Model

The AI Model page brings the machine learning models from Chapter 4 directly into the dashboard: the user sets six planet properties using live slicers, and a Python visual runs both trained models on the fly and renders the result in the same visual style as the rest of the report.

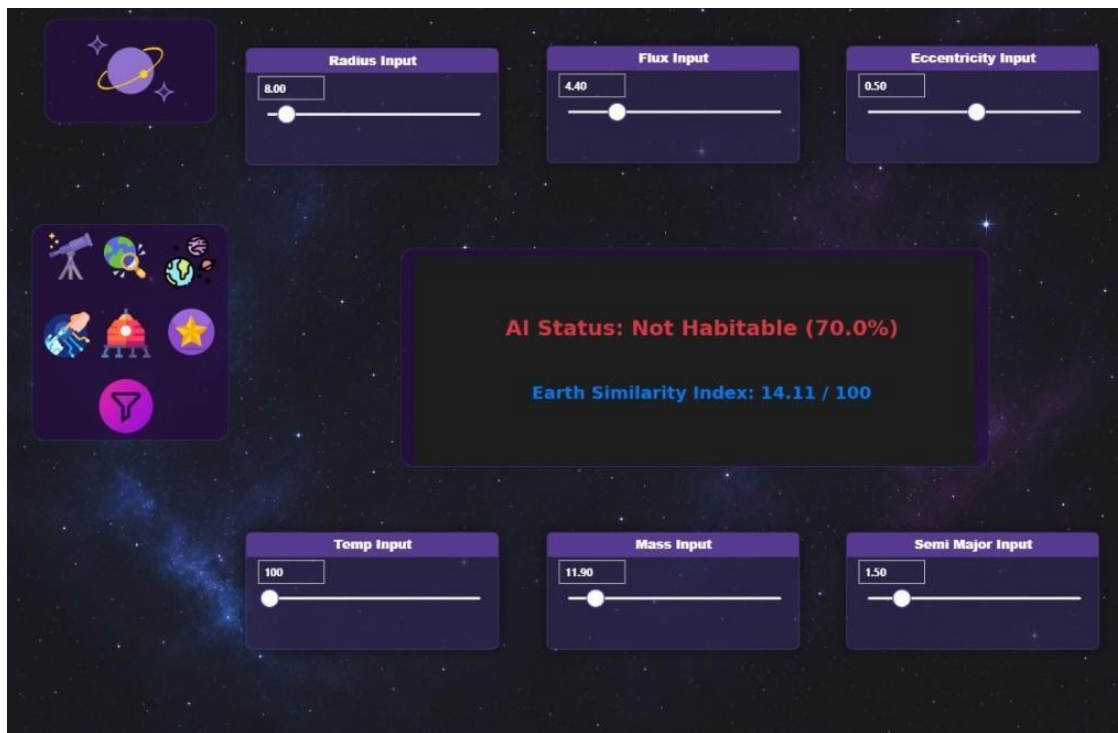


Figure 5.7 — AI Model Page: six live input slicers + classifier & ESI regressor output, rendered via a Python visual

Element	Type	Key Insight
Planet Radius	Slicer	User-adjustable input, Earth radii
Planet Mass	Slicer	User-adjustable input, Earth masses
Stellar Flux(K)	Slicer	User-adjustable input feeding the classifier/regressor
Planet Temperature(K)	Slicer	User-adjustable input, equilibrium temperature
Planet Semi Major Axis	Slicer	User-adjustable input, AU
Planet Eccentricity	Slicer	User-adjustable input, orbital eccentricity
AI Status & Earth Similarity Index	Python Visual	Loads habitability_classifier.pkl, feature_scaler.pkl, and earth_similarity_regressor.json; scales the six slicer values and renders “AI Status: Habitable / Not Habitable (confidence %)” and “Earth Similarity Index: X / 100” as a styled matplotlib figure matching the dashboard's dark theme (green for Habitable, red for Not Habitable, blue for the ESI line)

The Python visual's logic mirrors the Streamlit app from Chapter 4 exactly — same scaler, same two models, same six features — so the two interfaces (Power BI page and Streamlit app) are expected to return matching predictions for the same inputs.

6.5 Design Summary

Design Dimension	Choice Made	Rationale
Background Theme (all pages)	Deep space-black with a soft purple/violet nebula wash and scattered stars	Reinforces the astronomical subject matter across every page
Color Palette	Indigo & violet purples for KPI cards and chart series; warm gold accent for callouts and branding	Consistent, high-contrast "cosmic" identity distinct from a typical corporate BI palette
Sidebar Iconography	Fixed set of line-art icons (planet, telescope, globe, DNA helix, rocket, star, filter) repeated on every analytical page	Anchors navigation and reinforces the project's data-meets-discovery theme
Orbit / Atom Motif	Circular ring-of-icons diagram on Cover & Description pages	Visually represents the six analytical dimensions before the user reaches any data
KPI Cards	Four headline KPIs per analytical page, each with a themed icon	Scannable, executive-level insight at a glance, consistent with the reference documentation style
Interactivity	Four slicers per analytical page (mix of range sliders and dropdowns)	Lets users narrow the exploration without leaving the page's core narrative
Page Count	7 pages (Cover + Description + 4 analytical pages + 1 planned AI page)	Balances a complete data story with a clear, page-by-page narrative arc